

Quark — systematic trading research, done honestly

Savitur Swarup · github.com/Savi-Swar/StrategyBuilder

all results out-of-sample 2012+, net of costs, purged walk-forward, deflated for trial count

Headline finding

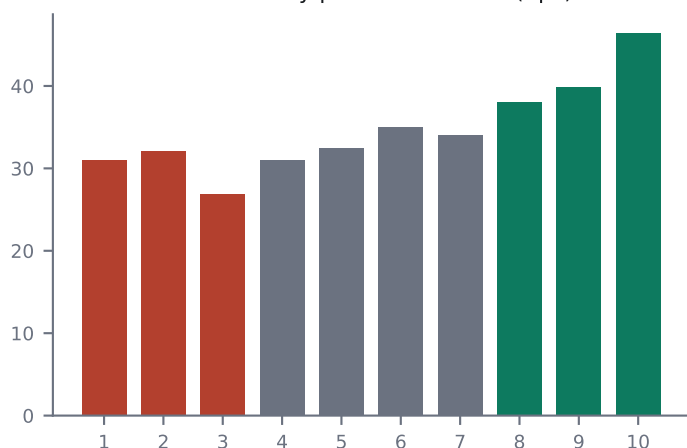
A gradient-boosted cross-sectional model over ~500 S&P 500 stocks predicts next-week relative returns: weekly rank IC = +0.0172 ($t = 3.35$, $n = 756$ non-overlapping weeks), near-monotonic decile spread. Economics are thin after costs — the honest headline is predictive power, not easy money:

net Sharpe 0.04 weekly, 0.26 monthly;

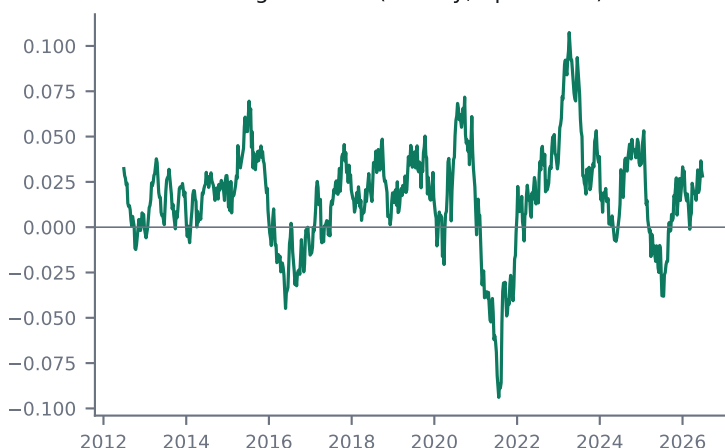
a pre-registered no-trade band (hold until rank decays 0.30) lifts it to 0.18 at half the cost drag.

Controls: shuffled-label AUC 0.500 / IC ≈ 0 ; the long side carries the signal (shorting did not pay).

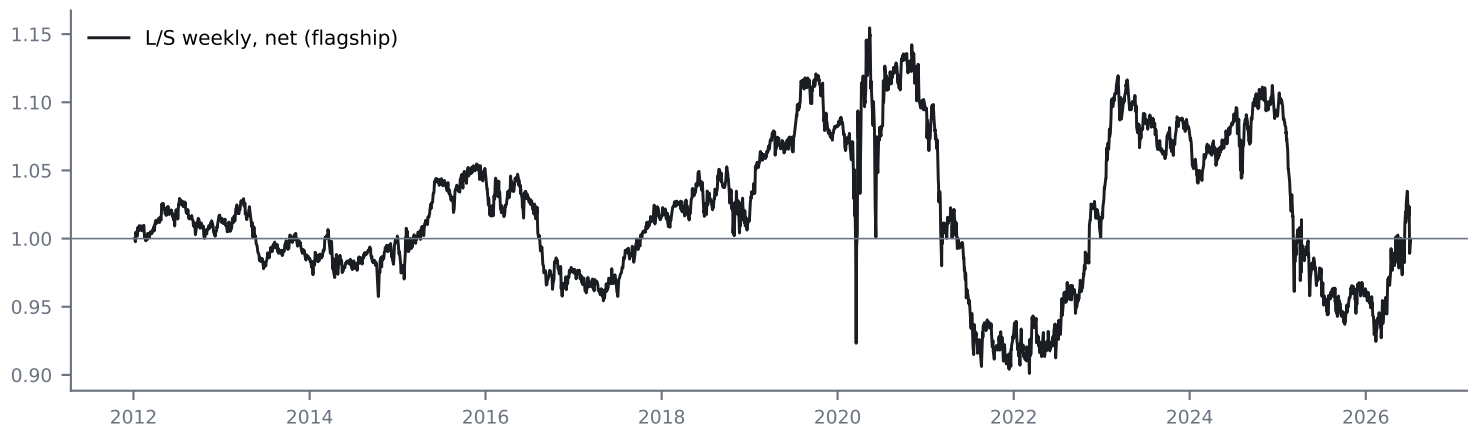
Gross fwd 5d return by prediction decile (bps)



26-week rolling mean IC (weekly, Spearman)



Dollar-neutral extreme-decile long/short — net equity curve



What failed, reported anyway

Classic indicators (8 counted variants): best = `ma_cross_50_200`, Deflated Sharpe Ratio 0.19 — the best of 8 coin flips. Daily ML timing on 43 multi-asset instruments: AUC 0.518 does not clear costs

(net Sharpe -0.44 , DSR ≈ 0). Survivorship bias measured, not just disclaimed: a point-in-time rerun

keeps the signal (IC +0.0124, $t = 2.35$) but halves the decile spread — shipped numbers are upper bounds.

Why the numbers can be trusted

Lookahead

positions = `signal.shift(lag)`, lag ≥ 1 enforced; unit test: prescient signal earns exactly zero

Leakage

purged + embargoed walk-forward (purge $\geq$ horizon); shuffled-label control run for every ML study

Costs

per-asset-class bps on turnover, entry bar charged; a missing cost rate raises, never defaults to free

Selection

every variant in a counted registry; best-of-family Sharpe deflated (Bailey & López de Prado DSR)

Verification

64 unit tests on synthetic fixtures; independent cross-source price verification in the daily job

Ships as “Vig”: a serverless self-grading terminal — every prediction logged to a ledger and scored after its horizon completes; a live trust gate turns the desk off when the trailing IC decays.